



Vector Databases and Unstructured Data Search

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Speaker



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Milvus: The most widely-adopted vector database

Milvus is an **Open-Source Vector Database** to **store, index, manage**, and **use** the massive number of **embedding vectors** generated by deep neural networks and LLMs.



267+

contributors



27K+

stars



11M+

docker pulls

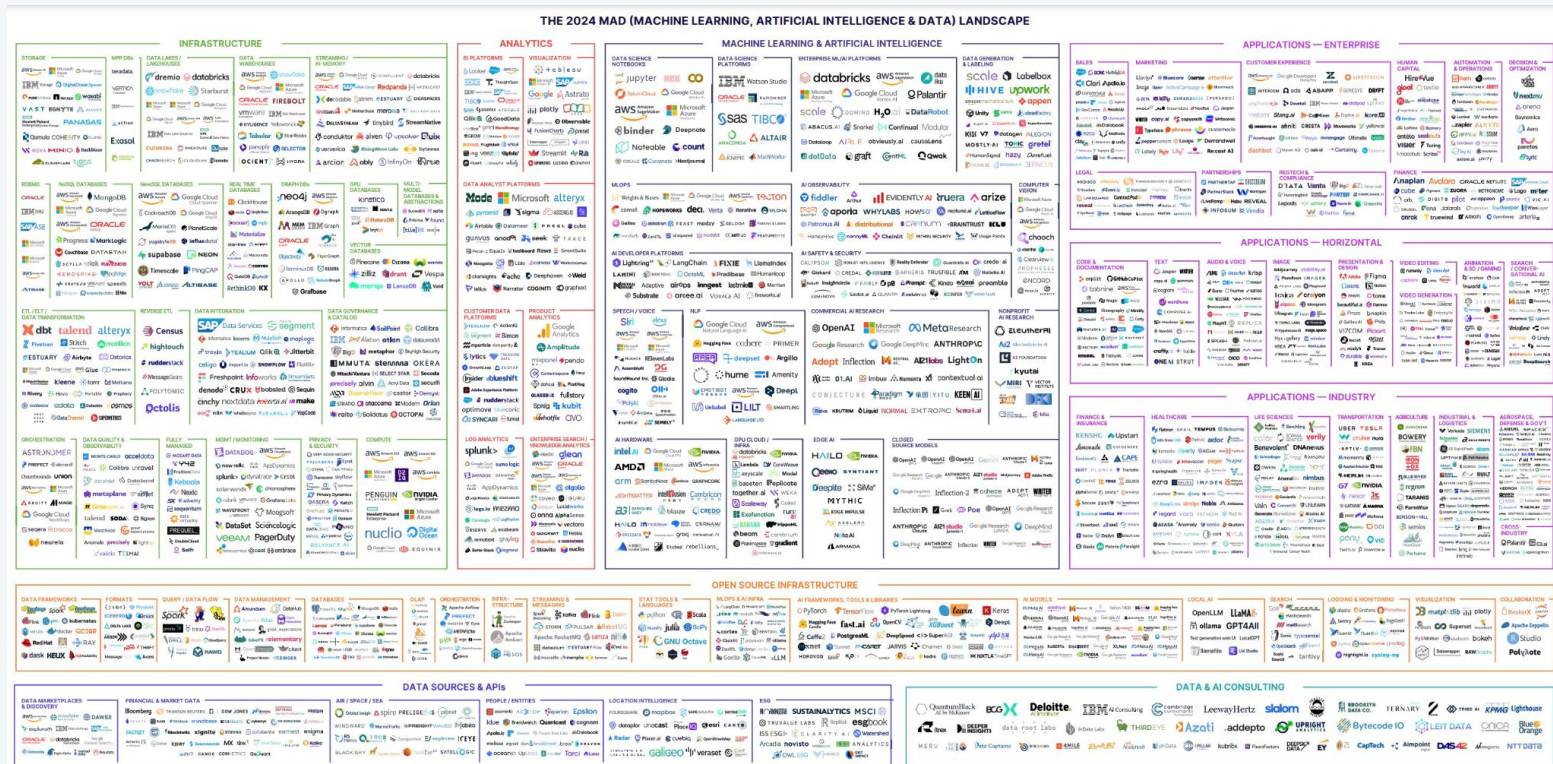


2K+

forks

The GenAI Landscape

It's a jungle out there



Version 1.0 - March 2024

© Matt Turck (@mattturck), Aman Kabber (@AmanKabber11) & FirstMark (@firstmarkcap)

Blog post: mattturck.com/MAD2024

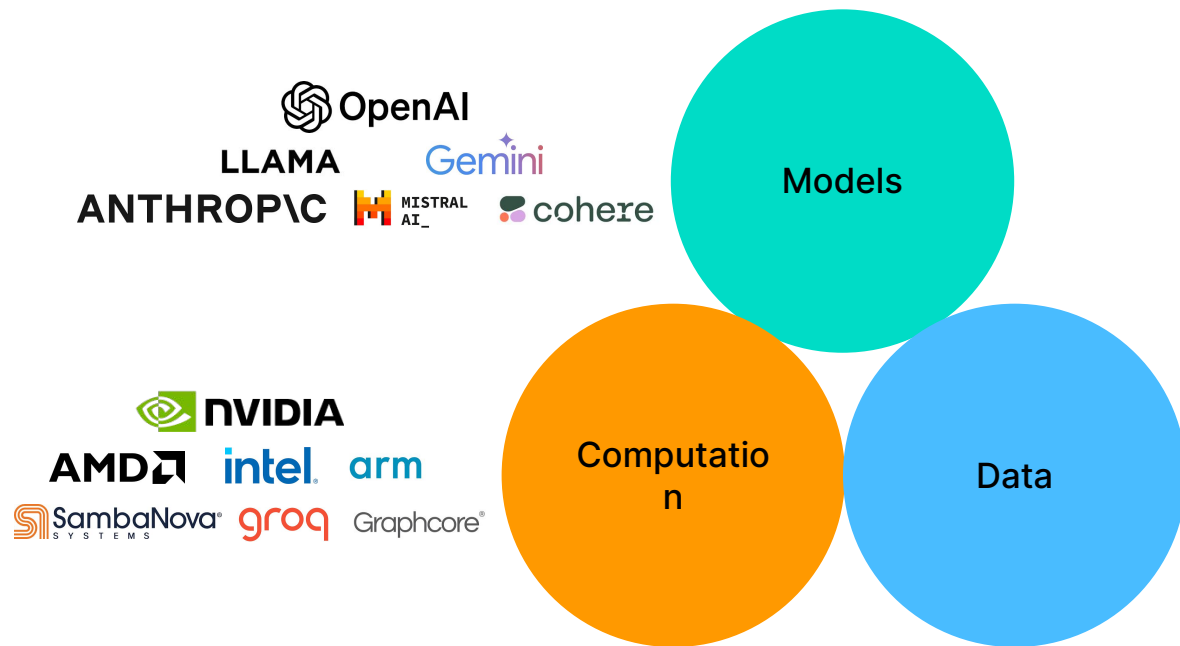
Interactive version: MAD.firstmarkcap.com

Comments? Email MAD2024@firstmarkcap.com

FIRSTMARK
CAPITAL PARTNERS



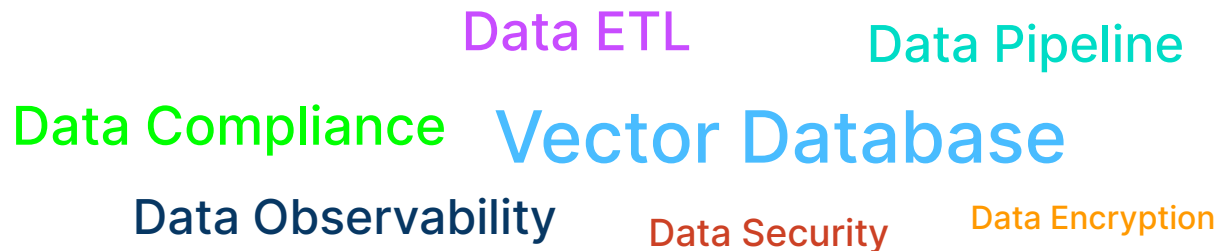
Three Pillars of GenAI Revisited



Opportunities in Unstructured Data

Vector Database

Opportunities in Unstructured Data



Unstructured Data Meetups

Our Mission

Foster **innovations**
and **collaborations** in
unstructured data

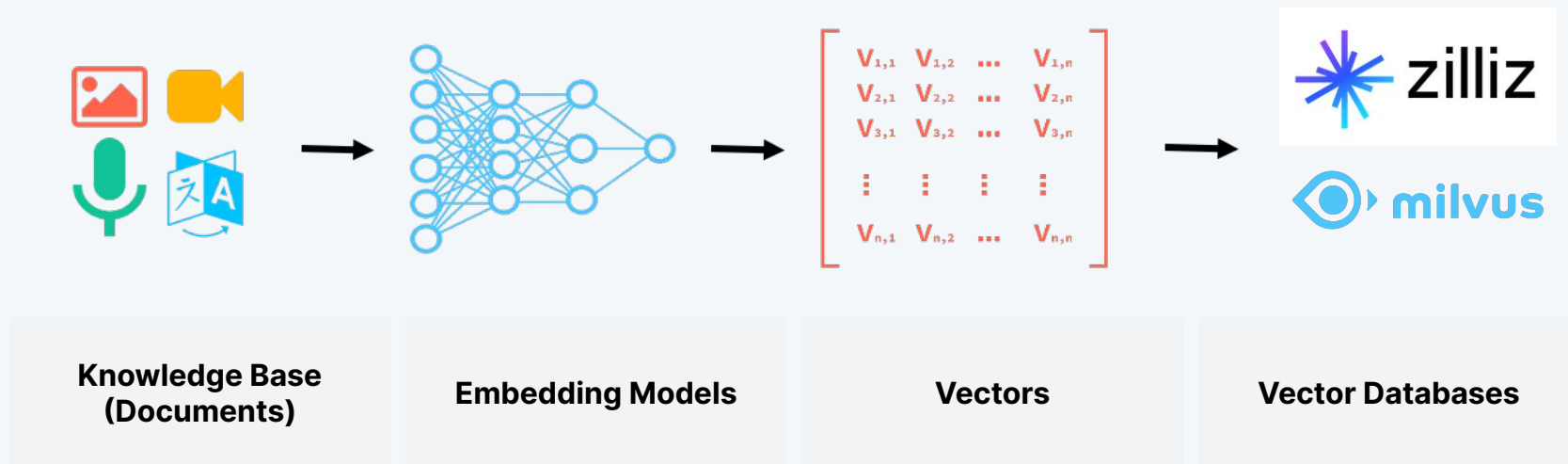
Vector Embedding Primer

Vector embeddings are something computers can understand



```
0.62 0.57 0.97 0.83 0.21 0.71 0.99 0.98 0.19 0.3 0.03 0.78
0.48 0.77 0.06 0.2 0.42 0.34 0.88 0.29 0.35 0.96 0.37 0.11
0.57 0.93 0.99 0.14 0.64 0.21 0.57 0.14 0.96 0.39 0.51 0.29
0.52 0.51 0.07 0.26 0.77 0.96 0.55 0.19 0.25 0.39 0.63 0.32
0.75 0.97 0.35 0.7 0.39 0.41 0.32 0.81 0.59 0.11 0.23 0.4
0.89 0.28 0.09 0.15 0.8 0.16 0.57 0.7 0.38 0.44 0.35 0.75
0.42 0.96 0.33 0.53 0.65 0.89 0.63 0.19 0.49 0.93 0.09 0.92
0.6 0.86 0.86 0.49 0.41 0.39 0.02 0.39 0.62 0.41 0.91 0.29
0.64 0.22 0.89 0.44 0.47 0.41 0.72 0.34 0.16 0.12 0.68 0.89
0.93 0.27 0.16 0.37 0.66 0.76 0.23 0.6 0.19 0.85 0.63 0.8
0.81 0.93 0.06 0.57 0.33 0.45 0.01 0.53 0.73 0.56 0.98 0.14
0.62 0.75 0.81 0.55 0.17 0.55 0.74 0.18 0.26 0.29 0.22 0.35
0.82 0.72 0.09 0.3 0.01 0.3 0.68 0.4 0.72 0.17 0.1 0.19
0.32 0.79 0.08 0.2 0.44 0.63 0.35 0.16 0.27 0.62 0.96 0.88
0.89 0.22 0.26 0.71 0.96 0.46 1 0.98 0.43 0.53 0.78 0.17
0.07 0.45 0.89 0.68 0.68 0.4 0.02 0.39 0.05 0.56 0.57 0.92
```

Vectors unlock unstructured data...



...and power a variety of different use cases



Retrieval Augmented Generation (RAG)

Expand LLMs' knowledge by incorporating external data sources into LLMs and your AI applications.



Recommender System

Match user behavior or content features with other similar ones to make effective recommendations.



Text/ Semantic Search

Search for semantically similar texts across vast amounts of natural language documents.



Image Similarity Search

Identify and search for visually similar images or objects from a vast collection of image libraries.



Video Similarity Search

Search for similar videos, scenes, or objects from extensive collections of video libraries.



Audio Similarity Search

Find similar audios in large datasets for tasks like genre classification or speech recognition



Molecular Similarity Search

Search for similar substructures, superstructures, and other structures for a specific molecule.



Anomaly Detection

Detect data points, events, and observations that deviate significantly from the usual pattern



Multimodal Similarity Search

Search over multiple types of data simultaneously, e.g. text and images

History of Embeddings

Embeddings models workhorses of AI apps

 **Hugging Face**

 **sentence-transformers** **all-MiniLM-L6-v2**  like 1.32k

 Sentence Similarity  Sentence Transformers  PyTorch  TensorFlow  Rust  s2orc

 flax-sentence-embeddings/stackexchange_xml  ms_marco  gooaq  yahoo_answers_topics

 code_search_net  search_qa  eli5  snli  multi_nli  wikihow  natural_questions

 trivia_qa  embedding-data/sentence-compression  embedding-data/flickr30k-captions

 embedding-data/altlex  embedding-data/simple-wiki  embedding-data/QQP

 embedding-data/SPECTER  embedding-data/PAQ_pairs  embedding-data/WikiAnswers  English

 bert  feature-extraction  Inference Endpoints  arxiv:1904.06472  arxiv:2102.07033

 arxiv:2104.08727  arxiv:1704.05179  arxiv:1810.09305  License: apache-2.0

 Deploy  Use in sentence-transformers

 **Model card**  Files  Community **42**

 Edit model card

all-MiniLM-L6-v2

This is a [sentence-transformers](#) model: It maps

 Downloads last month **5,523,635** 

 **Hugging Face**

 **meta-llama** **Llama-2-7b-chat-hf**  like 2.5k

 Text Generation  Transformers  PyTorch  Safetensors  English  llama  facebook

 meta  llama-2  text-generation-inference  arxiv:2307.09288

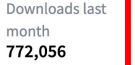
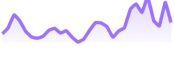
 Train  Deploy  Use in Transformers

 **Model card**  Files

 **Access Llama 2 on Hugging Face**

This is a form to enable access to Llama 2 on Hugging Face after you have been granted access from Meta. Please visit the [Meta website](#) and accept our license terms and acceptable use policy before submitting this form. Requests will be processed in 1-2 days.

Your Hugging Face account email address MUST match the email you provide on the Meta website, or your request will not be approved.

 Downloads last month **772,056** 

 **Safetensors**

 Model size **6.74B params**

 Tensor type **FP16** 

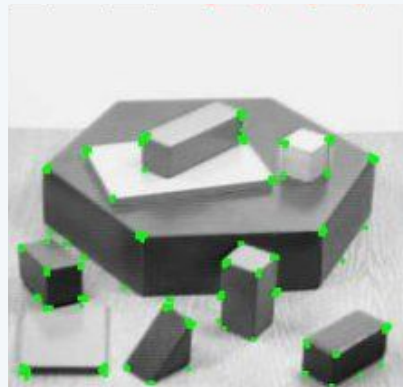
 Text Generation

Inference API has been turned off for this

Back in the day...



$$w_{x,y} = tf_{x,y} \times \log \left(\frac{N}{df_x} \right)$$

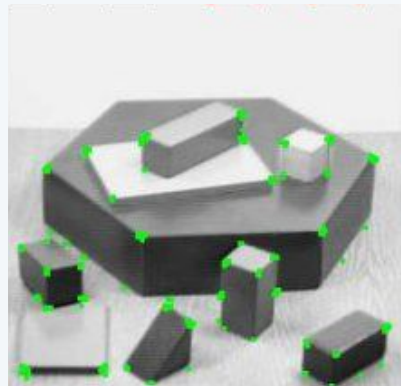


Back in the day...

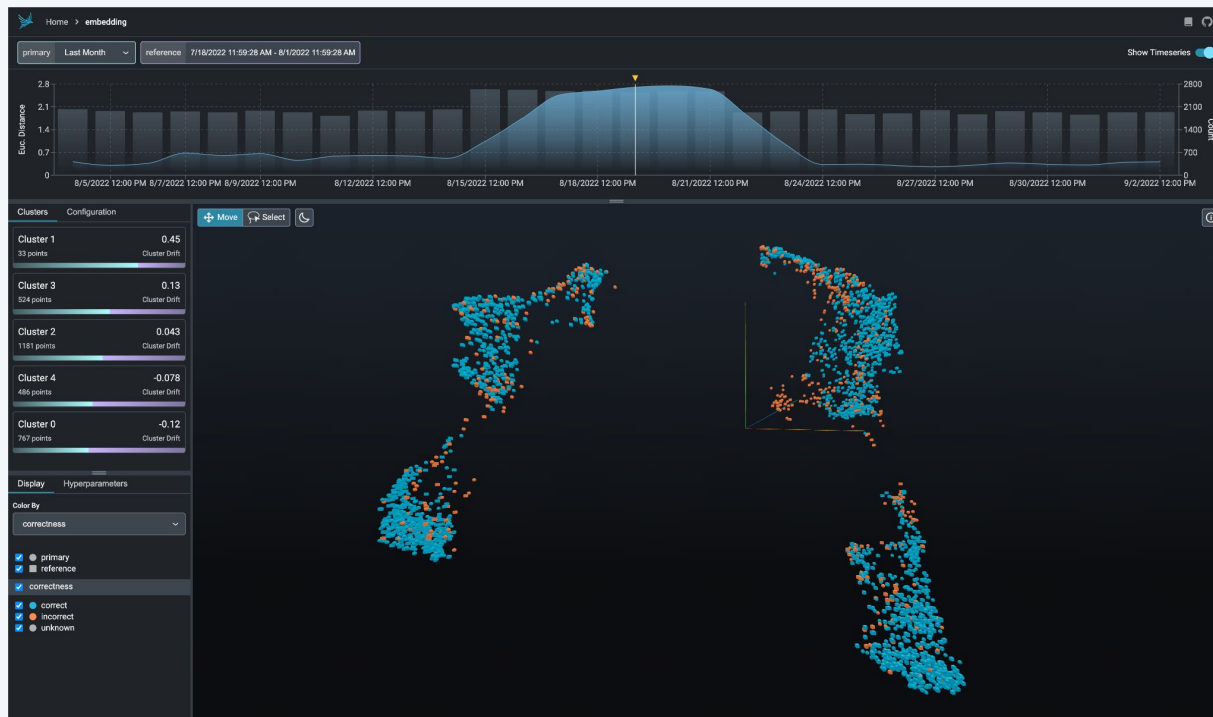


$$w_{x,y} = tf_{x,y} \times \log \left(\frac{N}{df_x} \right)$$

“Handcrafted features”



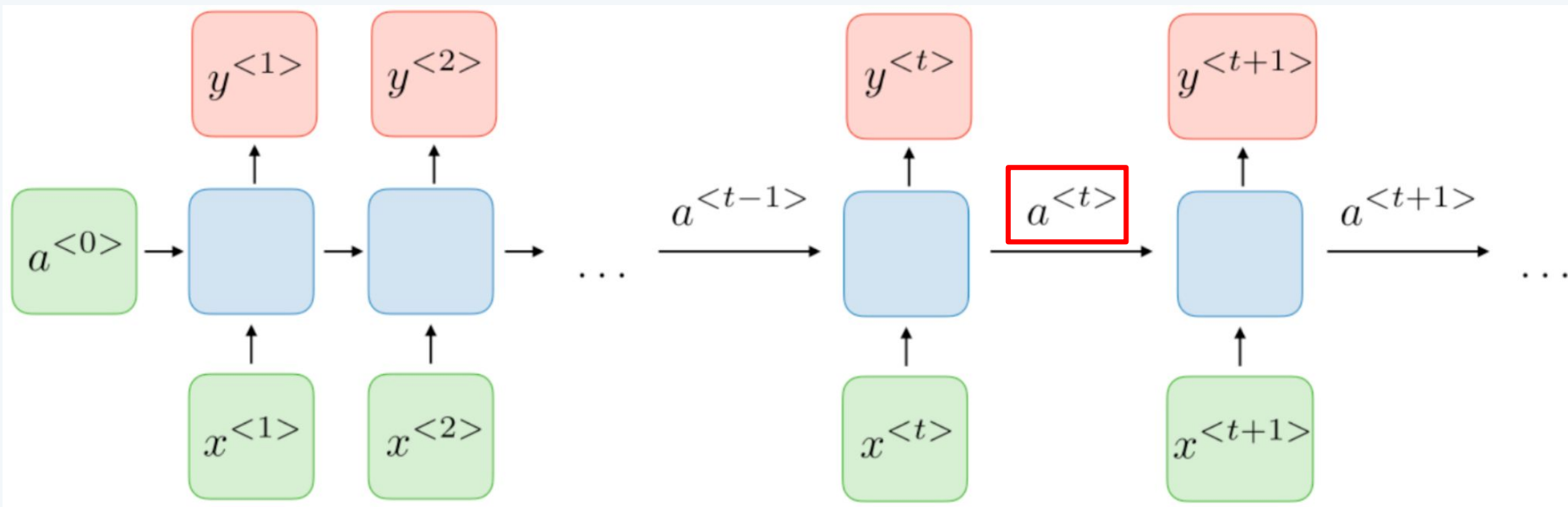
Modern day embeddings



Source: [Arize Phoenix](#)

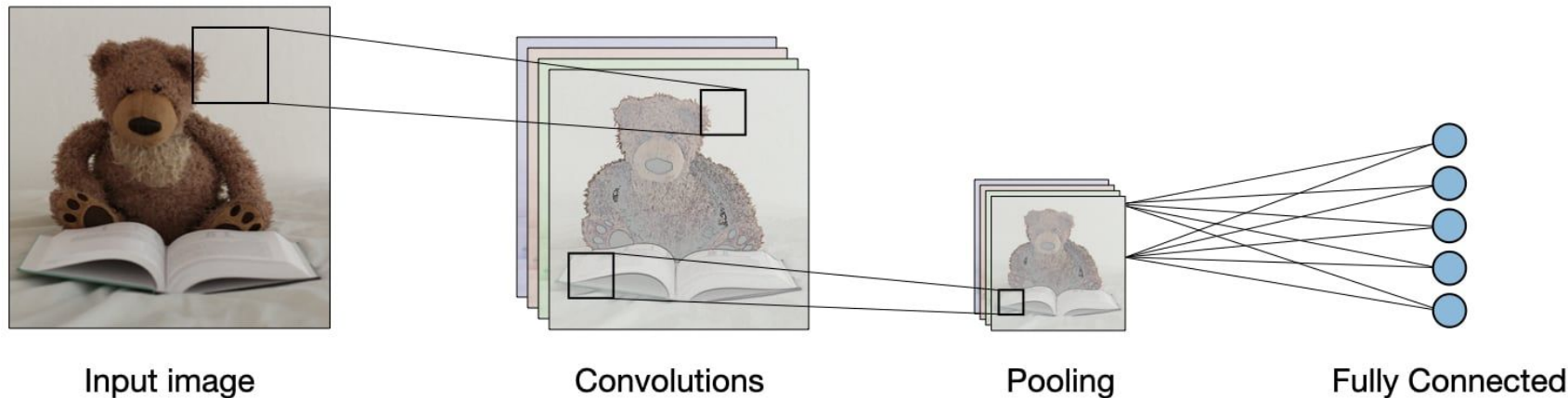
Embedding Models

Recurrent neural networks



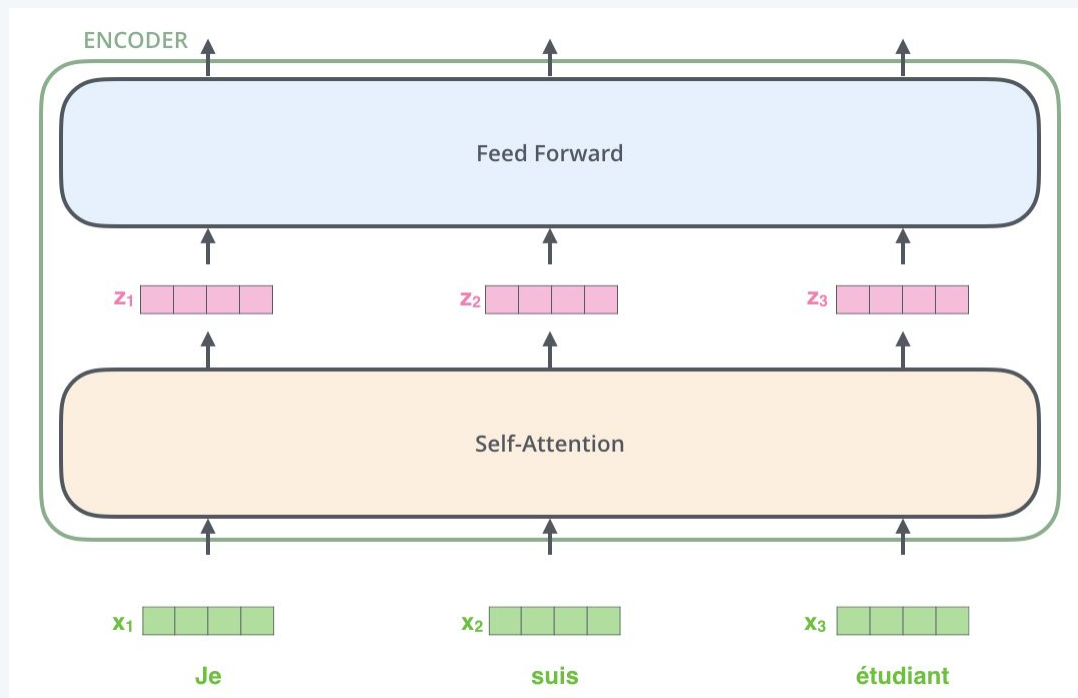
Source: [CS230 notes](#)

Convolutional networks



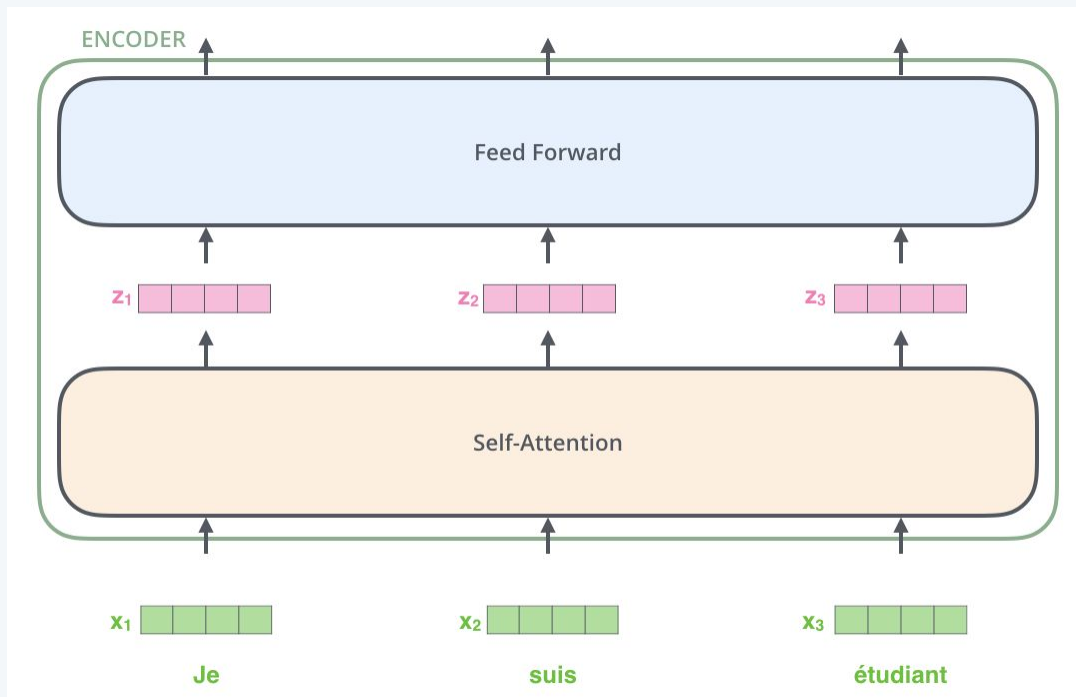
Source: [CS230 notes](#)

Transformer encoder



Source: [Illustrated Transformer](#)

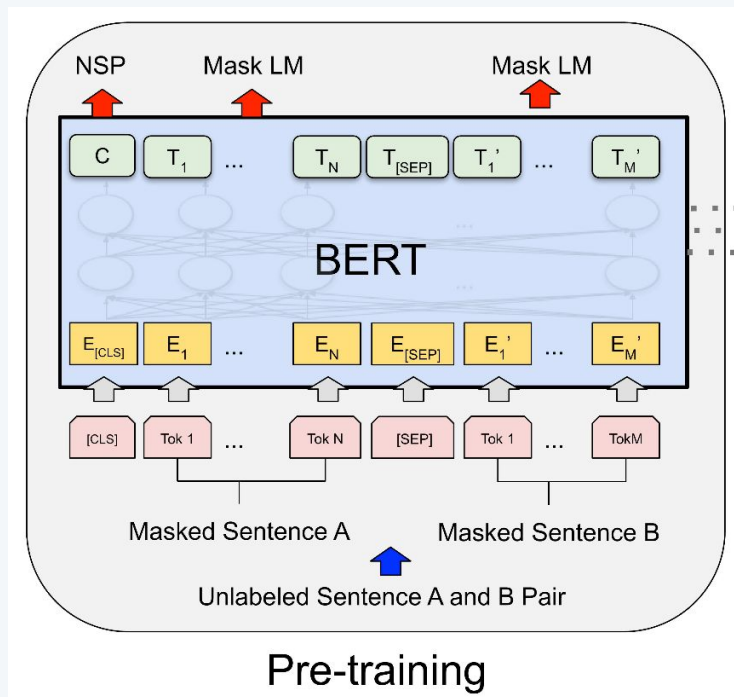
Transformer encoder



One
embedding
per token!

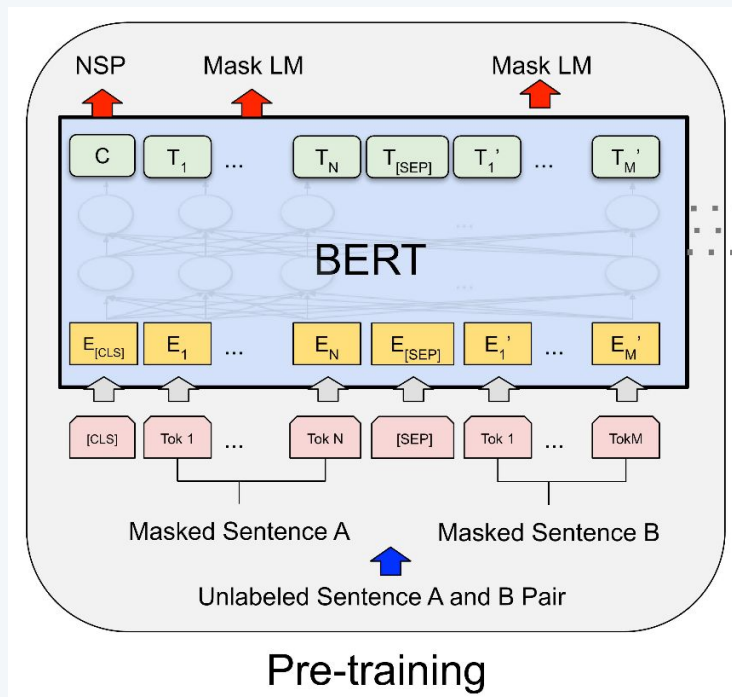
Source: [Illustrated Transformer](#)

BERT



Source: [BERT paper](#)

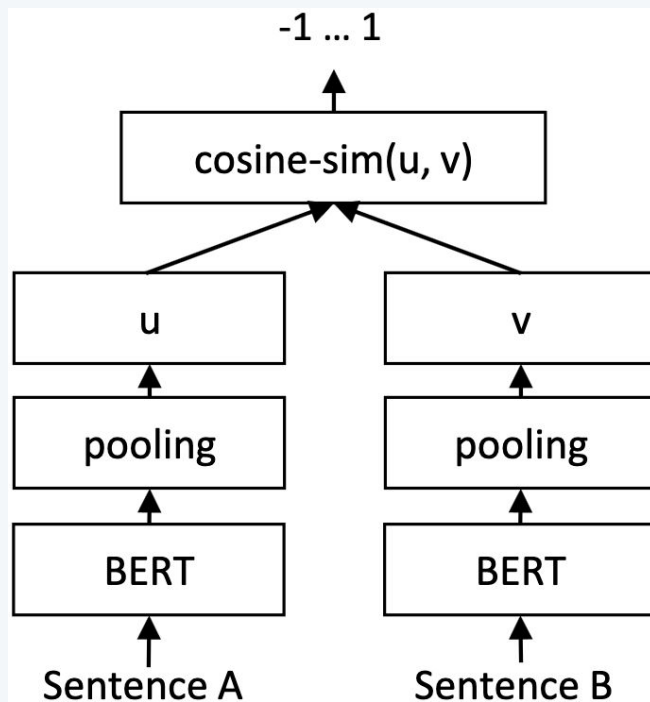
BERT



Source: [BERT paper](#)

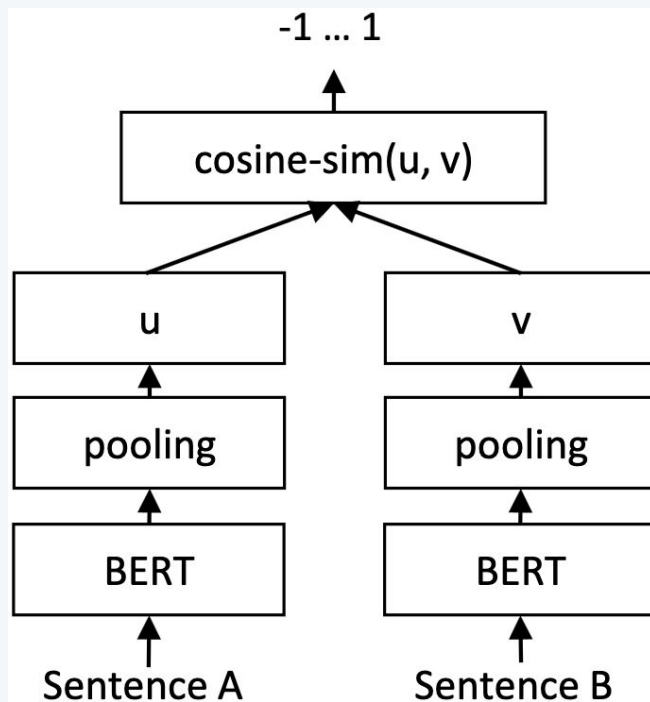
Cross-encoder
(one inference per
query/document
pair)

Sentence BERT



Source: [SBERT paper](#)

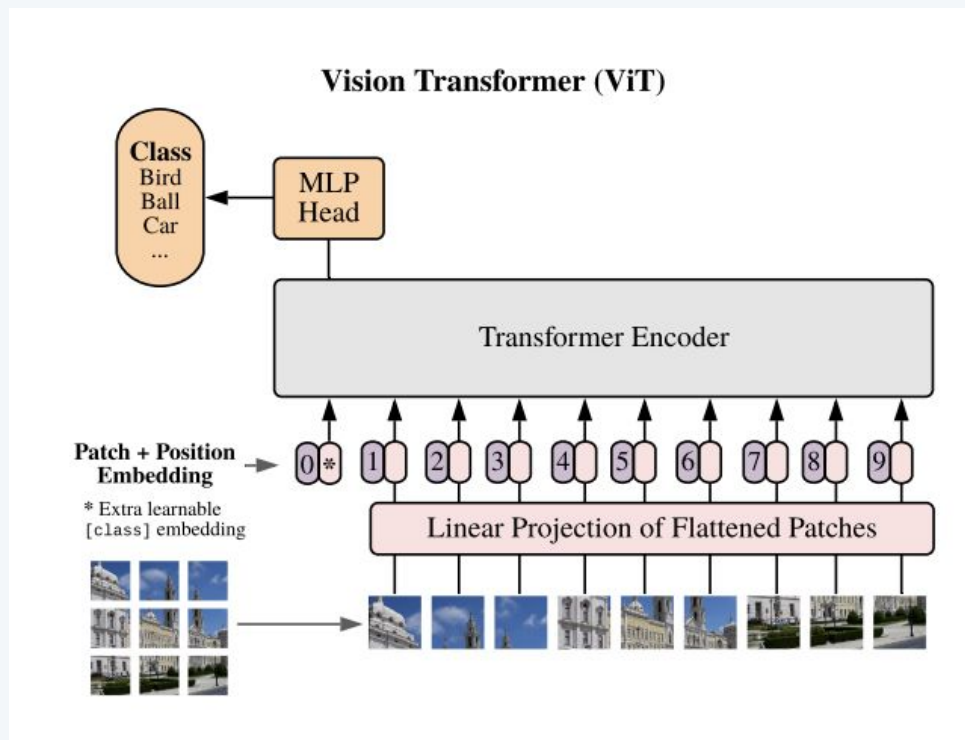
Sentence BERT



Source: [SBERT paper](#)

Bi-encoder
1 embedding per text

Vision transformer



Source: [ViT paper](#)

Embedding models today

- **Text embeddings: some flavor of SBERT**
 - Self-supervised pre-training (masking + NSP)
 - Contrastive (regression) or triplet loss
 - Changes to model arch or training, e.g. sparse attention, masking entities, or a different objective function
- **Image embeddings: hybrid convnet/self-attention**
 - Trained across a large labelled or weakly labelled dataset
 - Categorical cross-entropy or binary cross-entropy loss

Bonus: Multimodal RAG

Multi-modal Retrieval

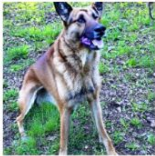
1) Cross-Modal Retrieval

Audio	Images & Videos	Depth	Text
 Crackle of a Fire			"A fire crackles while a pan of food is frying on the fire." "Fire is crackling then wind starts blowing." "Firewood crackles then music..."
 Baby Cooing			"A baby is crying while a toddler is laughing." "A baby is laughing while an adult is laughing." "A baby laughs and something..."

2) Embedding-Space Arithmetic

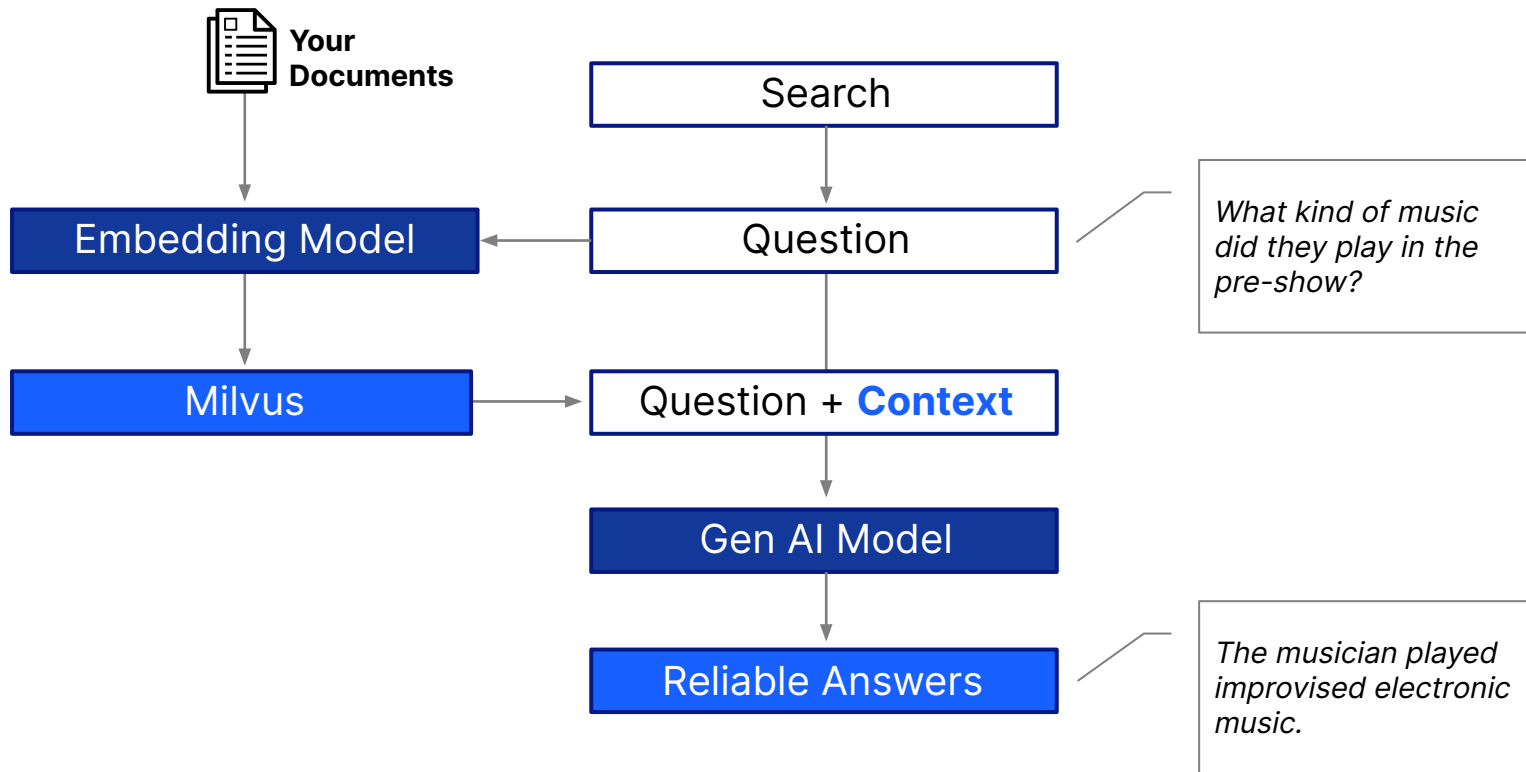
 $\oplus$ 
Waves $\rightarrow$ 

3) Audio to Image Generation

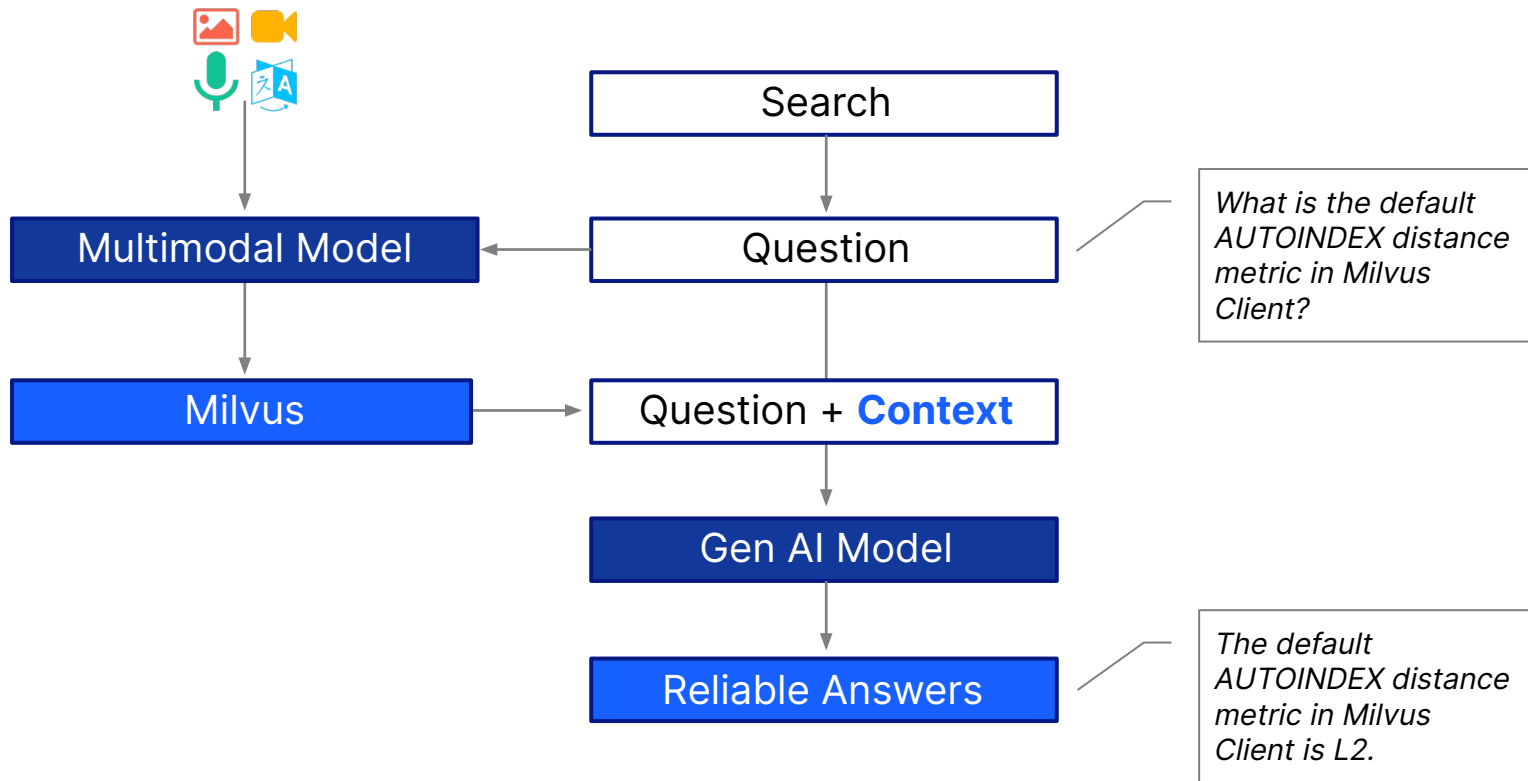

Dog
Engine
Fire
Rain

[Girdhar, et al.](#)

Vanilla RAG (RAG 1.0)



RAG 2.0



Demo Time

THANK YOU



We need your stars!

<https://github.com/milvus-io/milvus>

 **Join our discord:** <https://discord.gg/FjCMmaJng6>

